Precalculus — Algebra I

Version 4 $- \epsilon$

by

Carl Stitz, Ph.D.
Lakeland Community College

Jeff Zeager, Ph.D.
Lorain County Community College

$\sigma o \phi \acute{\imath} \alpha$

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Preface

Thank you for your interest in our book, but more importantly, thank you for taking the time to read the Preface. I always read the Prefaces of the textbooks which I use in my classes because I believe it is in the Preface where I begin to understand the authors - who they are, what their motivation for writing the book was, and what they hope the reader will get out of reading the text. Pedagogical issues such as content organization and how professors and students should best use a book can usually be gleaned out of its Table of Contents, but the reasons behind the choices authors make should be shared in the Preface. Also, I feel that the Preface of a textbook should demonstrate the authors' love of their discipline and passion for teaching, so that I come away believing that they really want to help students and not just make money. Thus, I thank my fellow Preface-readers again for giving me the opportunity to share with you the need and vision which guided the creation of this book and passion which both Carl and I hold for Mathematics and the teaching of it.

Carl and I are natives of Northeast Ohio. We met in graduate school at Kent State University in 1997. I finished my Ph.D in Pure Mathematics in August 1998 and started teaching at Lorain County Community College in Elyria, Ohio just two days after graduation. Carl earned his Ph.D in Pure Mathematics in August 2000 and started teaching at Lakeland Community College in Kirtland, Ohio that same month. Our schools are fairly similar in size and mission and each serves a similar population of students. The students range in age from about 16 (Ohio has a Post-Secondary Enrollment Option program which allows high school students to take college

courses for free while still in high school.) to over 65. Many of the "non-traditional" students are returning to school in order to change careers. A majority of the students at both schools receive some sort of financial aid, be it scholarships from the schools' foundations, state-funded grants or federal financial aid like student loans, and many of them have lives busied by family and job demands. Some will be taking their Associate degrees and entering (or re-entering) the workforce while others will be continuing on to a four-year college or university. Despite their many differences, our students share one common attribute: they do not want to spend $200 on a College Algebra book.

The challenge of reducing the cost of textbooks is one that many states, including Ohio, are taking quite seriously. Indeed, state-level leaders have started to work with faculty from several of the colleges and universities in Ohio and with the major publishers as well. That process will take considerable time so Carl and I came up with a plan of our own. We decided that the best way to help our students right now was to write our own College Algebra book and give it away electronically for free. We were granted sabbaticals from our respective institutions for the Spring semester of 2009 and actually began writing the textbook on December 16, 2008. Using an open-source text editor called TexNicCenter and an open-source distribution of LaTeX called MikTex 2.7, Carl and I wrote and edited all of the text, exercises and answers and created all of the graphs (using Metapost within LaTeX) for Version $0.\overline{9}$ in about eight months. (We choose to create a text in only black and white to keep printing costs to a minimum for those students who prefer a printed edition. This somewhat Spartan page layout stands in sharp relief to the explosion of colors found in most other College Algebra texts, but neither Carl nor I believe the four-color print adds anything of value.) I used the book in three sections of College Algebra at Lorain County Community College in the Fall of 2009 and Carl's colleague, Dr. Bill Previts, taught a section of College Algebra at Lakeland with the book that semester as well. Students had the option of downloading the book as a .pdf file from our website www.stitz-zeager.com[1] or buying a low-cost printed version from our colleges' respective bookstores. (By

[1] http://www.stitz-zeager.com

giving this book away for free electronically, we end the cycle of new editions appearing every 18 months to curtail the used book market.) During Thanksgiving break in November 2009, many additional exercises written by Dr. Previts were added and the typographical errors found by our students and others were corrected. On December 10, 2009, Version $\sqrt{2}$ was released. The book remains free for download at our website and by using Lulu.com[2] as an on-demand printing service, our bookstores are now able to provide a printed edition for just under $19. Neither Carl nor I have, or will ever, receive any royalties from the printed editions. As a contribution back to the open-source community, all of the LaTeX files used to compile the book are available for free under a Creative Commons License on our website as well. That way, anyone who would like to rearrange or edit the content for their classes can do so as long as it remains free.

The only disadvantage to not working for a publisher is that we don't have a paid editorial staff. What we have instead, beyond ourselves, is friends, colleagues and unknown people in the open-source community who alert us to errors they find as they read the textbook. What we gain in not having to report to a publisher so dramatically outweighs the lack of the paid staff that we have turned down every offer to publish our book. (As of the writing of this Preface, we've had three offers.) By maintaining this book by ourselves, Carl and I retain all creative control and keep the book our own. We control the organization, depth and rigor of the content which means we can resist the pressure to diminish the rigor and homogenize the content so as to appeal to a mass market. A casual glance through the Table of Contents of most of the major publishers' College Algebra books reveals nearly isomorphic content in both order and depth. Our Table of Contents shows a different approach, one that might be labeled "Functions First." To truly use The Rule of Four, that is, in order to discuss each new concept algebraically, graphically, numerically and verbally, it seems completely obvious to us that one would need to introduce functions first. (Take a moment and compare our ordering to the classic "equations first, then the Cartesian Plane and THEN functions" approach seen in most of

[2] http://www.lulu.com/content/paperback-book/college-algebra/7513097

the major players.) We then introduce a class of functions and discuss the equations, inequalities (with a heavy emphasis on sign diagrams) and applications which involve functions in that class. The material is presented at a level that definitely prepares a student for Calculus while giving them relevant Mathematics which can be used in other classes as well. Graphing calculators are used sparingly and only as a tool to enhance the Mathematics, not to replace it. The answers to nearly all of the computational homework exercises are given in the text and we have gone to great lengths to write some very thought provoking discussion questions whose answers are not given. One will notice that our exercise sets are much shorter than the traditional sets of nearly 100 "drill and kill" questions which build skill devoid of understanding. Our experience has been that students can do about 15-20 homework exercises a night so we very carefully chose smaller sets of questions which cover all of the necessary skills and get the students thinking more deeply about the Mathematics involved.

Critics of the Open Educational Resource movement might quip that "open-source is where bad content goes to die," to which I say this: take a serious look at what we offer our students. Look through a few sections to see if what we've written is bad content in your opinion. I see this open-source book not as something which is "free and worth every penny", but rather, as a high quality alternative to the business as usual of the textbook industry and I hope that you agree. If you have any comments, questions or concerns please feel free to contact me at jeff@stitz-zeager.com or Carl at carl@stitz-zeager.com.

Jeff Zeager
Lorain County Community College
January 25, 2010

Chapter 1

Introduction to Functions

1.1 Functions and their Representations

1.2 Constant and Linear Functions

1.3 Absolute Value Functions

1.4 Quadratic Functions

Chapter 2

Polynomial Functions

2.1 Graphs of Polynomial Functions

2.2 The Remainder and Factor Theorems

In Section **??** we saw how much of the 'local' behavior of the graph of a polynomial function is determined by the zeros of the polynomial function. In that section, the polynomial functions we were given were mostly, if not completely, factored which greatly simplified the process for determining zeros. In this section, we revisit the relationship between zeros and factors with the ultimate aim of taking a polynomial function given to us in the form stated in Definition **??** and determining its zeros.

We start by way of example: suppose we wish to determine the zeros of $f(x) = x^3+4x^2-5x-14$. Setting $f(x) = 0$ results in the polynomial equation $x^3 + 4x^2 - 5x - 14 = 0$. Despite all of the factoring techniques we learned (and forgot!) in Intermediate Algebra, this equation foils[1] us at every turn. Knowing that the zeros of f correspond to x-intercepts on the graph of $y = f(x)$, we use a graphing utility to produce the graph in Figure 2.2.1. The graph suggests that the function has three zeros, one of which appears to be $x = 2$ and two others for whom we are provided what we assume to be decimal approximations: $x \approx -4.414$ and $x \approx -1.586$. We can verify if these are zeros easily enough. We find $f(2) = (2)^2 + 4(2)^2 - 5(2) - 14 = 0$, but $f(-4.414) \approx 0.0039$ and $f(-1.586) \approx 0.0022$, While these last two values are probably by some measures, 'close' to 0, they are not *exactly* equal to 0. The question becomes: is there a way to use the fact that $x = 2$ is a zero to obtain the other two zeros? Based on our experience, if $x = 2$ is a zero, it seems that there should be a factor of $(x - 2)$ lurking around in the factorization of $f(x)$. In other words, we should expect that $x^3 + 4x^2 - 5x - 14 = (x - 2)\,q(x)$, where $q(x)$ is some other polynomial. How could we find such a $q(x)$, if it even exists? The answer comes from our old friend, polynomial division. (See Section **??**.) Below, we perform the long division: $(x^3 + 4x^2 - 5x - 14) \div (x - 2)$ and obtain $x^2 + 6x + 7$.

[1] pun intended

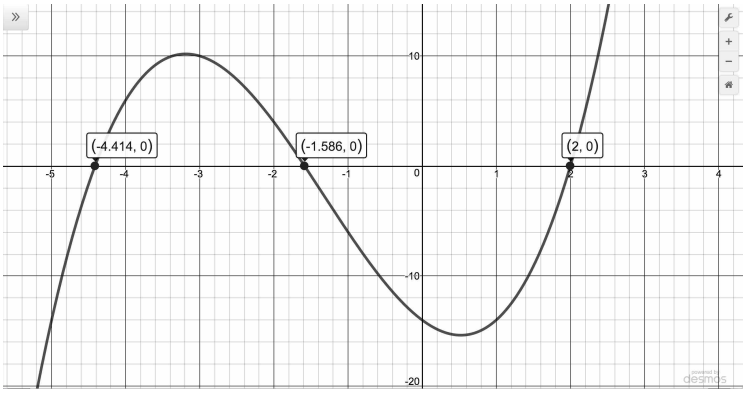

Figure 2.2.1

$$
\begin{array}{r|rrrr}
 & & x^2 & +\ 6x & +\ 7 \\
\hline
x-2 & x^3 & +\ 4x^2 & -\ 5x & -\ 14 \\
 & -\left(x^3\right. & \left.-\ 2x^2\right) & & \\
\cline{2-3}
 & & 6x^2 & -\ 5x & \\
 & & -\left(6x^2\right. & \left.-\ 12x\right) & \\
\cline{3-4}
 & & & 7x & -\ 14 \\
 & & & -\ (7x & -\ 14) \\
\cline{4-5}
 & & & & 0
\end{array}
$$

Said differently, $f(x) = x^3 + 4x^2 - 5x - 14 = (x - 2)\left(x^2 + 6x + 7\right)$. Using this form of $f(x)$, we find the zeros by solving $(x - 2)\left(x^2 + 6x + 7\right) = 0$. Setting each factor equal to 0, we get $x - 2 = 0$ (which gives us our known zero, $x = 2$) as well as $x^2 + 6x + 7 = 0$. The latter doesn't factor nicely, so we apply the Quadratic Formula to get $x = -3 \pm \sqrt{2}$. Sure enough, $-3 - \sqrt{2} \approx -4.414$ and $-3 + \sqrt{2} \approx -1.586$. We leave it to the reader to show $f(-3 - \sqrt{2}) = 0$ and $f(-3 + \sqrt{2}) = 0$. (See Exercise 36.)

The point of this section is to generalize the technique applied here. First up is a friendly reminder of what we can expect when we divide polynomials.

Theorem 2.2.1. Polynomial Division:
Suppose $d(x)$ and $p(x)$ are nonzero polynomial functions where the degree of p is greater than or equal to the degree of d. There exist two unique polynomial functions, $q(x)$ and $r(x)$, such that $p(x) = d(x)\,q(x) + r(x)$, where either $r(x) = 0$ or the degree of r is strictly less than the degree of d.

As you may recall, all of the polynomials in Theorem 2.2.1 have special names. The polynomial p is called the **dividend**; d is the **divisor**; q is the **quotient**; r is the **remainder**. If $r(x) = 0$ then d is called a **factor** of p. The word 'unique' here is critical in that it guarantees there is only *one* quotient and remainder for each division problem.[2] The proof of Theorem 2.2.1 is usually relegated to a course in Abstract Algebra, but we can still use the result to establish two important facts which are the basis of the rest of the chapter.

Theorem 2.2.2. The Remainder Theorem: Suppose p is a polynomial function of degree at least 1 and c is a real number. When $p(x)$ is divided by $x - c$ the remainder is $p(c)$. Said differently, there is a polynomial function $q(x)$ such that:

$$p(x) = (x - c)q(x) + p(c)$$

The proof of Theorem 2.2.2 is a direct consequence of Theorem 2.2.1. Since $x - c$ has degree 1, when a polynomial function is divided by $x - c$, the remainder is either 0 or degree 0 (i.e., a nonzero constant.) In either case, $p(x) = (x - c)\,q(x) + r$, where r, the remainder, is a real number, possibly 0. It follows that $p(c) = (c - c)\,q(c) + r = 0 \cdot q(c) + r = r$, so we get $r = p(c)$ as required. There is one last 'low hanging fruit'[3] to collect which we present below.

[2] Hence the use of the definite article 'the' when speaking of *the* quotient and *the* remainder.

[3] Jeff hates this expression and Carl included it just to annoy him.

Theorem 2.2.3. The Factor Theorem:
Suppose p is a nonzero polynomial function. The real number c is a zero of p if and only if $(x - c)$ is a factor of $p(x)$.

Once again, we see the phrase 'if and only if' which means there are really two things being said in The Factor Theorem: if $(x - c)$ is a factor of $p(x)$, then c is a zero of p and the *only* way c is a zero of p is if $(x - c)$ is a factor of $p(x)$. We argue the Factor Theorem as follows: if $(x - c)$ is a factor of $p(x)$, then $p(x) = (x - c)\,q(x)$ for some polynomial q. Hence, $p(c) = (c - c)\,q(c) = 0$, so c is a zero of p. Conversely, suppose c is a zero of p, so $p(c) = 0$. The Remainder Theorem tells us $p(x) = (x-c)q(x)+p(c) = (x-c)q(x)+0 = (x-c)q(x)$. Hence, $(x-c)$ is a factor of $p(x)$.

We have enough theory to explain why the concept of multiplicity (Definition **??**) is well-defined. If c is a zero of p, then The Factor Theorem tells us there is a polynomial function q_1 so that $p(x) = (x - c)q_1(x)$. If $q_1(c) = 0$, then we apply the Factor Theorem to q_1 and find a polynomial q_2 so that $q_1(x) = (x - c)q_2(x)$. Hence, we have

$$p(x) = (x - c)q_1(x) = (x - c)(x - c)q_2(x) = (x - c)^2 q_2(x).$$

We now 'rinse and repeat' this process. Since the degree of p is a finite number, this process has to end at some point. That is we arrive at a factorization $p(x) = (x - c)^m q(x)$ where $q(c) \neq 0$. Suppose we arrive at a different factorization of p using other methods. That is, we find $p(x) = (x - c)^k Q(x)$, where Q is a polynomial function with $Q(c) \neq 0$. Then we have $(x - c)^m q(x) = (x - c)^k Q(x)$. If $m \neq k$, then either $m < k$ or $m > k$. Assuming the former, then we may divide both sides by $(x - c)^m$ to get: $q(x) = (x - c)^{k-m} Q(x)$. Since $k > m$, $k - m > 0$ and we would have $q(c) = (c - c)^{k-m} Q(c) = 0$, a contradiction since we are assuming $q(c) \neq 0$. The assumption that $m > k$ likewise ends in a contradiction. Therefore, we have $m = k$, so $p(x) = (x - c)^m q(x) = (x - c)^m Q(x)$. By the uniqueness guaranteed in Theorem 2.2.1, we must have that $q(x) = Q(x)$. Hence, we have shown the number m, as well as the quotient polynomial

$q(x)$ are unique. The process outlined above, in which we coax out factors of $p(x)$ one at a time until we have all of them serves as a template for our work to come.

Of the things The Factor Theorem tells us, the most pragmatic is that we had better find a more efficient way to divide polynomial functions by quantities of the form $x - c$. Fortunately, people like Ruffini[4] and Horner[5] have already blazed this trail. Let's take a closer look at the long division we performed at the beginning of the section and try to streamline it. First off, let's change all of the subtractions into additions by distributing through the -1s.

$$
\begin{array}{r|rrrr}
 & & x^2 & +\ 6x & +\ 7 \\
\hline
x-2 & x^3 & +\ 4x^2 & -\ 5x & -14 \\
 & -x^3 & +\ 2x^2 & & \\
\cline{2-3}
 & & 6x^2 & -\ 5x & \\
 & & -6x^2 & +\ 12x & \\
\cline{3-4}
 & & & 7x & -14 \\
 & & & -7x & +14 \\
\cline{4-5}
 & & & & 0
\end{array}
$$

Next, observe that the terms $-x^3$, $-6x^2$ and $-7x$ are the exact opposite of the terms above them. The algorithm we use ensures this is always the case, so we can omit them without losing any information. Also note that the terms we 'bring down' (namely the $-5x$ and -14) aren't really necessary to recopy, so we omit them, too.

[4] http://en.wikipedia.org/wiki/Synthetic_division
[5] http://en.wikipedia.org/wiki/Horner_scheme

$$
\begin{array}{r|rrrr}
 & & x^2 & +\ 6x & +\ 7 \\
\hline
x-2 & x^3 & +4x^2 & -\ 5x & -14 \\
 & & 2x^2 & & \\
\cline{2-3}
 & & 6x^2 & & \\
 & & & 12x & \\
\cline{3-4}
 & & & 7x & \\
 & & & & 14 \\
\cline{4-5}
 & & & & 0
\end{array}
$$

Let's move terms up a bit and copy the x^3 into the last row.

$$
\begin{array}{r|rrrr}
 & & x^2 & +\ 6x & +\ 7 \\
\hline
x-2 & x^3 & +4x^2 & -\ 5x & -14 \\
\hline
 & & 2x^2 & 12x & 14 \\
\hline
 & x^3 & 6x^2 & 7x & 0
\end{array}
$$

Note that by arranging things in this manner, each term in the last row is obtained by adding the two terms above it. Notice also that the quotient polynomial can be obtained by dividing each of the first three terms in the last row by x and adding the results. If you take the time to work back through the original division problem, you will find that this is exactly the way we determined the quotient polynomial. This means that we no longer need to write the quotient polynomial down, nor the x in the divisor, to determine our answer.

$$
\begin{array}{r|rrrr}
-2 & x^3 & +4x^2 & -\ 5x & -14 \\
\hline
 & & 2x^2 & 12x & 14 \\
\hline
 & x^3 & 6x^2 & 7x & 0
\end{array}
$$

We've streamlined things quite a bit so far, but we can still do more. Let's take a moment to remind ourselves where the $2x^2$, $12x$ and 14 came from

in the second row. Each of these terms was obtained by multiplying the terms in the quotient, x^2, $6x$ and 7, respectively, by the -2 in $x-2$, then by -1 when we changed the subtraction to addition. Multiplying by -2 then by -1 is the same as multiplying by 2, so we replace the -2 in the divisor by 2. Furthermore, the coefficients of the quotient polynomial match the coefficients of the first three terms in the last row, so we now take the plunge and write only the coefficients of the terms to get

$$\begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & & 2 & 12 & 14 \\ \hline & 1 & 6 & 7 & 0 \end{array}$$

We have constructed a **synthetic division tableau** for this polynomial division problem. Let's re-work our division problem using this tableau to see how it greatly streamlines the division process. To divide $x^3+4x^2-5x-14$ by $x-2$, we write 2 in the place of the divisor and the coefficients of $x^3+4x^2-5x-14$ in for the dividend. Then 'bring down' the first coefficient of the dividend.

$$\begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ \hline \end{array} \qquad \begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & & & \\ \hline & 1 & & & \end{array}$$

Next, take the 2 from the divisor and multiply by the 1 that was 'brought down' to get 2. Write this underneath the 4, then add to get 6.

$$\begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & 2 & & \\ \hline & 1 & & & \end{array} \qquad \begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & 2 & & \\ \hline & 1 & 6 & & \end{array}$$

Now take the 2 from the divisor times the 6 to get 12, and add it to the -5 to get 7.

$$\begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & 2 & 12 & \\ \hline & 1 & 6 & & \end{array} \qquad \begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & 2 & 12 & \\ \hline & 1 & 6 & 7 & \end{array}$$

Finally, take the 2 in the divisor times the 7 to get 14, and add it to the -14 to get 0.

$$\begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & 2 & 12 & 14 \\ \hline & 1 & 6 & 7 & \end{array} \qquad \begin{array}{r|rrrr} 2 & 1 & 4 & -5 & -14 \\ & \downarrow & 2 & 12 & 14 \\ \hline & 1 & 6 & 7 & \boxed{0} \end{array}$$

The first three numbers in the last row of our tableau are the coefficients of the quotient polynomial. Remember, we started with a third degree polynomial and divided by a first degree polynomial, so the quotient is a second degree polynomial. Hence the quotient is $x^2 + 6x + 7$. The number in the box is the remainder. Synthetic division is our tool of choice for dividing polynomials by divisors of the form $x - c$. It is important to note that it works *only* for these kinds of divisors.[6] Also take note that when a polynomial (of degree at least 1) is divided by $x - c$, the result will be a polynomial of exactly one less degree. Finally, it is worth the time to trace each step in synthetic division back to its corresponding step in long division. While the authors have done their best to indicate where the algorithm comes from, there is no substitute for working through it yourself.

Example 2.2.1. Use synthetic division to perform the following polynomial divisions. Identify the quotient and remainder. Write the dividend, quotient and remainder in the form given in Theorem 2.2.1.

1. $\left(5x^3 - 2x^2 + 1\right) \div (x - 3)$
2. $\left(t^3 + 8\right) \div (t + 2)$
3. $\dfrac{4 - 8z - 12z^2}{2z - 3}$

Solution.

[6] You'll need to use good old-fashioned polynomial long division for divisors of degree larger than 1.

1. When setting up the synthetic division tableau, the coefficients of even 'missing' terms need to be accounted for, so we enter 0 for the coefficient of x in the dividend.

$$\begin{array}{r|rrrr} 3 & 5 & -2 & 0 & 1 \\ & \downarrow & 15 & 39 & 117 \\ \hline & 5 & 13 & 39 & \boxed{118} \end{array}$$

   Since the dividend was a third degree polynomial function, the quotient is a second degree (quadratic) polynomial function with coefficients 5, 13 and 39: $q(x) = 5x^2 + 13x + 39$. The remainder is $r(x) = 118$. According to Theorem 2.2.1, we have $5x^3 - 2x^2 + 1 = (x-3)\left(5x^2 + 13x + 39\right) + 118$, which we leave to the reader to check.

2. To use synthetic division here, we rewrite $t + 2$ as $t - (-2)$ and proceed as before

$$\begin{array}{r|rrrr} -2 & 1 & 0 & 0 & 8 \\ & \downarrow & -2 & 4 & -8 \\ \hline & 1 & -2 & 4 & \boxed{0} \end{array}$$

   We get the quotient $q(t) = t^2 - 2t + 4$ and the remainder $r(t) = 0$. Relating the dividend, quotient and remainder gives: $t^3 + 8 = (t + 2)\left(t^2 - 2t + 4\right)$, which is a specific instance of the 'sum of cubes' formula some of you may recall from Intermediate Algebra.

3. To divide $4 - 8z - 12z^2$ by $2z - 3$, two things must be done. First, we write the dividend in descending powers of z as $-12z^2 - 8z + 4$. Second, since synthetic division works only for factors of the form $z - c$, we factor $2z - 3$ as $2\left(z - \frac{3}{2}\right)$. Hence, we are dividing $-12z^2 - 8z + 4$ by two factors: 2 and $\left(z - \frac{3}{2}\right)$. Dividing first by 2, we obtain $-6z^2 - 4z + 2$. Next, we divide $-6z^2 - 4z + 2$ by $\left(z - \frac{3}{2}\right)$:

$$\begin{array}{c|ccc} \frac{3}{2} & -6 & -4 & 2 \\ & \downarrow & -9 & -\frac{39}{2} \\ \hline & -6 & -13 & \boxed{-\frac{35}{2}} \end{array}$$

Hence, $-6z^2 - 4z + 2 = \left(z - \frac{3}{2}\right)(-6z - 13) - \frac{35}{2}$. However when it comes to writing the dividend, quotient and remainder in the form given in Theorem 2.2.1, we need to find $q(z)$ and $r(z)$ so that $-12z^2 - 8z + 4 = (2z - 3)q(z) + r(z)$. Hence, starting with $-6z^2 - 4z + 2 = \left(z - \frac{3}{2}\right)(-6z - 13) - \frac{35}{2}$, we multiply 2 back on both sides:

$$\begin{aligned} -6z^2 - 4z + 2 &= \left(z - \tfrac{3}{2}\right)(-6z - 13) - \tfrac{35}{2} \\ 2\left(-6z^2 - 4z + 2\right) &= 2\left[\left(z - \tfrac{3}{2}\right)(-6z - 13) - \tfrac{35}{2}\right] \\ -12z^2 - 8z + 4 &= 2\left(z - \tfrac{3}{2}\right)(-6z - 13) - 2\left(\tfrac{35}{2}\right) \\ -12z^2 - 8z + 4 &= (2z - 3)(-6z - 13) - 35 \end{aligned}$$

At this stage, we have written $-12z^2 - 8z + 4$ in the **form** $(2z - 3)q(z) + r(z)$, so we identify the quotient as $q(z) = -6z - 13$ and the remainder is $r(z) = -35$. But how can we be sure these are the same quotient and remainder polynomial functions we would have obtained if we had taken the time to do the long division in the first place? Because of the word 'unique' in Theorem 2.2.1. The theorem states that there is only *one* way to decompose $-12z^2 - 8z + 4$ as $(2z - 3)q(z) + r(z)$. Since we have found such a way, we can be sure it is the only way.[7] □

The next example pulls together all of the concepts discussed in this section.

Example 2.2.2. Let $p(x) = 2x^3 - 5x + 3$.

1. Find $p(-2)$ using The Remainder Theorem. Check your answer by substitution.

[7] But it wouldn't hurt to check, just this once.

2. Verify $x = 1$ is a zero of p and use this information to all the real zeros of p.

Solution.

1. The Remainder Theorem states $p(-2)$ is the remainder when $p(x)$ is divided by $x - (-2)$. We set up our synthetic division tableau below. We are careful to record the coefficient of x^2 as 0:

$$\begin{array}{r|rrrr} -2 & 2 & 0 & -5 & 3 \\ & \downarrow & -4 & 8 & -6 \\ \hline & 2 & -4 & 3 & \boxed{-3} \end{array}$$

According to the Remainder Theorem, $p(-2) = -3$. We can check this by direct substitution into the formula for $p(x)$: $p(-2) = 2(-2)^3 - 5(-2) + 3 = -16 + 10 + 3 = -3$.

2. We verify $x = 1$ is a zero of p by evaluating $p(1) = 2(1)^3 - 5(1) + 3 = 0$. To see if there are any more real zeros, we need to solve $p(x) = 2x^3 - 5x + 3 = 0$. From the Factor Theorem, we know since $p(1) = 0$, we can factor $p(x)$ as $(x - 1)q(x)$. To find $q(x)$, we use synthetic division:

$$\begin{array}{r|rrrr} 1 & 2 & 0 & -5 & 3 \\ & \downarrow & 2 & 2 & -3 \\ \hline & 2 & 2 & -3 & \boxed{0} \end{array}$$

As promised, our remainder is 0, and we get $p(x) = (x-1)\left(2x^2 + 2x - 3\right)$. Setting this form of $p(x)$ equal to 0 we get $(x-1)\left(2x^2 + 2x - 3\right) = 0$. We recover $x = 1$ from setting $x - 1 = 0$ but we also obtain $x = \frac{-1 \pm \sqrt{7}}{2}$ from $2x^2 + 2x - 3 = 0$, courtesy of the Quadratic Formula. $\square$

Our next example demonstrates how we can extend the synthetic division tableau to accommodate zeros of multiplicity greater than 1.

Example 2.2.3. Let $p(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$. Show $x = \frac{1}{2}$ is a zero of multiplicity 2 and find all of the remaining real zeros of p.

Solution. While computing $p\left(\frac{1}{2}\right) = 0$ shows $x = \frac{1}{2}$ is a zero of p, to prove it has multiplicity 2, we need to factor $p(x) = \left(x - \frac{1}{2}\right)^2 q(x)$ with $q\left(\frac{1}{2}\right) \neq 0$,. We set up for synthetic division, but instead of stopping after the first division, we continue the tableau downwards and divide $\left(x - \frac{1}{2}\right)$ directly into the quotient we obtained from the first division as follows:

$$
\begin{array}{c|rrrrr}
\frac{1}{2} & 4 & -4 & -11 & 12 & -3 \\
 & \downarrow & 2 & -1 & -6 & 3 \\
\hline
\frac{1}{2} & 4 & -2 & -12 & 6 & \boxed{0} \\
 & \downarrow & 2 & 0 & -6 & \\
\hline
 & 4 & 0 & -12 & \boxed{0} &
\end{array}
$$

We get:[8] $4x^4 - 4x^3 - 11x^2 + 12x - 3 = \left(x - \frac{1}{2}\right)^2 \left(4x^2 - 12\right)$. Note if we let $q(x) = 4x^2 - 12$, then $q\left(\frac{1}{2}\right) = 4\left(\frac{1}{2}\right)^2 - 12 = -11 \neq 0$ which proves $x = \frac{1}{2}$ is a zero of p of multiplicity 2. To find the remaining zeros of p, we set the quotient $4x^2 - 12 = 0$, so $x^2 = 3$ and extract square roots to get $x = \pm\sqrt{3}$. $\square$

A couple of things about the last example are worth mentioning. First, the extension of the synthetic division tableau for repeated divisions will be a common site in the sections to come. Typically, we will start with a higher order polynomial and peel off one zero at a time until we are left with a quadratic, whose roots can always be found using the Quadratic Formula. Secondly, we found $x = \pm\sqrt{3}$ are zeros of p. The Factor Theorem guarantees $\left(x - \sqrt{3}\right)$ and $\left(x - \left(-\sqrt{3}\right)\right)$ are both factors of p. We can certainly put the Factor Theorem to the test and continue the synthetic division tableau from above to see what happens.

[8] For those wanting more detail: the first division gives: $4x^4 - 4x^3 - 11x^2 + 12x - 3 = \left(x - \frac{1}{2}\right)\left(4x^3 - 2x^2 - 12x + 6\right)$. The second division gives: $4x^3 - 2x^2 - 12x + 6 = \left(x - \frac{1}{2}\right)\left(4x^2 - 12\right)$.

$$
\begin{array}{r|rrrrr}
\frac{1}{2} & 4 & -4 & -11 & 12 & -3 \\
 & \downarrow & 2 & -1 & -6 & 3 \\
\hline
\frac{1}{2} & 4 & -2 & -12 & 6 & \boxed{0} \\
 & \downarrow & 2 & 0 & -6 & \\
\hline
\sqrt{3} & 4 & 0 & -12 & \boxed{0} & \\
 & \downarrow & 4\sqrt{3} & 12 & & \\
\hline
-\sqrt{3} & 4 & 4\sqrt{3} & \boxed{0} & & \\
 & \downarrow & -4\sqrt{3} & & & \\
\hline
 & 4 & \boxed{0} & & &
\end{array}
$$

This gives us

$$
\begin{aligned}
p(x) &= 4x^4 - 4x^3 - 11x^2 + 12x - 3 \\
&= \left(x - \tfrac{1}{2}\right)^2 \left(x - \sqrt{3}\right) \left(x - \left(-\sqrt{3}\right)\right) \quad (4) \\
&= 4\left(x - \tfrac{1}{2}\right)^2 \left(x - \sqrt{3}\right) \left(x - \left(-\sqrt{3}\right)\right)
\end{aligned}
$$

We have shown that p is a product of its leading coefficient times linear factors of the form $(x - c)$ where c are zeros of p. It may surprise and delight the reader that, in theory, all polynomials can be reduced to this kind of factorization. We leave that discussion to Section **??**, because the zeros may not be real numbers. Our final theorem in the section gives us an upper bound on the number of real zeros.

Theorem 2.2.4. Suppose f is a polynomial of degree $n \geq 1$. Then f has at most n real zeros, counting multiplicities.

Theorem 2.2.4 is a consequence of the Factor Theorem and polynomial multiplication. Every zero c of f gives us a factor of the form $(x - c)$ for $f(x)$. Since f has degree n, there can be at most n of these factors. The next section provides us some tools which not only help us determine where the real zeros are to be found, but which real numbers they may be.

We close this section with Box 2.2.1, containing a summary of several concepts previously presented. You should take the time to look back

through the text to see where each concept was first introduced and where each connection to the other concepts was made.

Box 2.2.1: Connections Between Zeros, Factors and Graphs of Polynomial Functions

Suppose p is a polynomial function of degree $n \geq 1$. The following statements are equivalent:

- The real number c is a zero of p
- $p(c) = 0$
- $x = c$ is a solution to the polynomial equation $p(x) = 0$
- $(x - c)$ is a factor of $p(x)$
- The point $(c, 0)$ is an x-intercept of the graph of $y = p(x)$

2.2.1 Exercises

In Exercises 1 - 14, use synthetic division to perform the following polynomial divisions. Identify the quotient and remainder. Write the divisor, quotient and remainder in the form given in Theorem 2.2.1.

1. $\left(3x^2 - 2x + 1\right) \div (x - 1)$
2. $\left(x^2 - 5\right) \div (x - 5)$
3. $\left(3 - 4t - 2t^2\right) \div (t + 1)$
4. $\left(4t^2 - 5t + 3\right) \div (t + 3)$
5. $\left(z^3 + 8\right) \div (z + 2)$
6. $\left(4z^3 + 2z - 3\right) \div (z - 3)$
7. $\left(18x^2 - 15x - 25\right) \div \left(x - \frac{5}{3}\right)$
8. $\left(4x^2 - 1\right) \div \left(x - \frac{1}{2}\right)$
9. $\left(2t^3 + t^2 + 2t + 1\right) \div \left(t + \frac{1}{2}\right)$
10. $\left(3t^3 - t + 4\right) \div \left(t - \frac{2}{3}\right)$
11. $\left(2z^3 - 3z + 1\right) \div \left(z - \frac{1}{2}\right)$
12. $\left(4z^4 - 12z^3 + 13z^2 - 12z + 9\right) \div \left(z - \frac{3}{2}\right)$
13. $\left(x^4 - 6x^2 + 9\right) \div \left(x - \sqrt{3}\right)$
14. $\left(x^6 - 6x^4 + 12x^2 - 8\right) \div \left(x + \sqrt{2}\right)$

In Exercises 15 - 24, find $p(c)$ using the Remainder Theorem. If $p(c) = 0$, use the Factor Theorem to partially factor the polynomial function.

15. $p(x) = 2x^2 - x + 1$, $c = 4$
16. $p(x) = 4x^2 - 33x - 180$, $c = 12$
17. $p(t) = 2t^3 - t + 6$, $c = -3$
18. $p(t) = t^3 + 2t^2 + 3t + 4$, $c = -1$
19. $p(z) = 3z^3 - 6z^2 + 4z - 8$, $c = 2$
20. $p(z) = 8z^3 + 12z^2 + 6z + 1$, $c = -\frac{1}{2}$
21. $p(x) = x^4 - 2x^2 + 4$, $c = \frac{3}{2}$
22. $p(x) = 6x^4 - x^2 + 2$, $c = -\frac{2}{3}$
23. $p(t) = t^4 + t^3 - 6t^2 - 7t - 7$, $c = -\sqrt{7}$
24. $p(t) = t^2 - 4t + 1$, $c = 2 - \sqrt{3}$

In Exercises 25 - 34, you are given a polynomial function and one of its zeros. Find the remaining real zeros and factor the polynomial.

25. $x^3 - 6x^2 + 11x - 6, \quad c = 1$
26. $x^3 - 24x^2 + 192x - 512, \quad c = 8$
27. $3t^3 + 4t^2 - t - 2, \quad c = \frac{2}{3}$
28. $2t^3 - 3t^2 - 11t + 6, \quad c = \frac{1}{2}$
29. $z^3 + 2z^2 - 3z - 6, \quad c = -2$
30. $2z^3 - z^2 - 10z + 5, \quad c = \frac{1}{2}$
31. $4x^4 - 28x^3 + 61x^2 - 42x + 9$, $c = \frac{1}{2}$ is a zero of multiplicity 2

32. $t^5 + 2t^4 - 12t^3 - 38t^2 - 37t - 12$, $c = -1$ is a zero of multiplicity 3
33. $125z^5 - 275z^4 - 2265z^3 - 3213z^2 - 1728z - 324$, $c = -\frac{3}{5}$ is a zero of multiplicity 3
34. $x^2 - 2x - 2, \quad c = 1 - \sqrt{3}$
35. Find a quadratic polynomial with integer coefficients which has $x = \dfrac{3}{5} \pm \dfrac{\sqrt{29}}{5}$ as its real zeros.
36. For $f(x) = x^3 + 4x^2 - 5x - 14$, show $f(-3 - \sqrt{2}) = 0$ and $f(-3 + \sqrt{2}) = 0$ two ways:
    (a) By direct substitution.
    (b) Using synthetic division and the Factor Theorem
37. Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + ... + a_2x^2 + a_1x + a_0$ be a polynomial function with the property that $a_n + a_{n-1} + ... + a_1 + a_0 = 0$. (That is, the sum of the coefficients and the constant term is 0.)

    Prove that $(x - 1)$ is a factor of $f(x)$.

    HINT: Show $f(1) = 0$ and invoke the Factor Theorem
38. Verify the result in number 37 with the functions: $f(x) = x^3 - 2x + 1$ and $f(x) = 3x^4 - x - 2$.
39. Suppose a is a nonzero real number. Find the quotients below, using synthetic division as required.
    - $\dfrac{x - a}{x - a}$
    - $\dfrac{x^2 - a^2}{x - a}$
    - $\dfrac{x^3 - a^3}{x - a}$
    - $\dfrac{x^4 - a^4}{x - a}$
    - $\dfrac{x^5 - a^5}{x - a}$

    Based on the pattern that evolves, find the quotient: $\dfrac{x^{10} - a^{10}}{x - a}$.

    What about $\dfrac{x^n - a^n}{x - a}$?
40. Use your result from number 39 to rewrite the sum: $1 + r + r^2 + \cdots + r^{n-2} + r^{n-1}$ as a quotient. What assumptions need to be made about r?

2.2.2 Answers

1. $\left(3x^2 - 2x + 1\right) = (x - 1)(3x + 1) + 2$
2. $\left(x^2 - 5\right) = (x - 5)(x + 5) + 20$
3. $\left(3 - 4t - 2t^2\right) = (t + 1)(-2t - 2) + 5$
4. $\left(4t^2 - 5t + 3\right) = (t + 3)(4t - 17) + 54$
5. $\left(z^3 + 8\right) = (z + 2)\left(z^2 - 2z + 4\right) + 0$
6. $\left(4z^3 + 2z - 3\right) = (z - 3)\left(4z^2 + 12z + 38\right) + 111$
7. $\left(18x^2 - 15x - 25\right) = \left(x - \frac{5}{3}\right)(18x + 15) + 0$
8. $\left(4x^2 - 1\right) = \left(x - \frac{1}{2}\right)(4x + 2) + 0$
9. $\left(2t^3 + t^2 + 2t + 1\right) = \left(t + \frac{1}{2}\right)\left(2t^2 + 2\right) + 0$
10. $\left(3t^3 - t + 4\right) = \left(t - \frac{2}{3}\right)\left(3t^2 + 2t + \frac{1}{3}\right) + \frac{38}{9}$
11. $\left(2z^3 - 3z + 1\right) = \left(z - \frac{1}{2}\right)\left(2z^2 + z - \frac{5}{2}\right) - \frac{1}{4}$
12. $\left(4z^4 - 12z^3 + 13z^2 - 12z + 9\right) = \left(z - \frac{3}{2}\right)\left(4z^3 - 6z^2 + 4z - 6\right) + 0$
13. $\left(x^4 - 6x^2 + 9\right) = \left(x - \sqrt{3}\right)\left(x^3 + \sqrt{3}\,x^2 - 3x - 3\sqrt{3}\right) + 0$
14. $\left(x^6 - 6x^4 + 12x^2 - 8\right) = \left(x + \sqrt{2}\right)\left(x^5 - \sqrt{2}\,x^4 - 4x^3 + 4\sqrt{2}\,x^2 + 4x - 4\sqrt{\right.$
    0
15. $p(4) = 29$
16. $p(12) = 0$, $p(x) = (x - 12)(4x + 15)$
17. $p(-3) = -45$
18. $p(-1) = 2$
19. $p(2) = 0$, $p(z) = (z - 2)\left(3z^2 + 4\right)$
20. $p\left(-\frac{1}{2}\right) = 0$, $p(z) = \left(z + \frac{1}{2}\right)\left(8z^2 + 8z + 2\right)$
21. $p\left(\frac{3}{2}\right) = \frac{73}{16}$
22. $p\left(-\frac{2}{3}\right) = \frac{74}{27}$
23. $p(-\sqrt{7}) = 0$, $p(t) = (t + \sqrt{7})\left(t^3 + (1 - \sqrt{7})t^2 + (1 - \sqrt{7})t - \sqrt{7}\right)$
24. $p(2 - \sqrt{3}) = 0$, $p(t) = (t - (2 - \sqrt{3}))(t - (2 + \sqrt{3}))$
25. $x^3 - 6x^2 + 11x - 6 = (x - 1)(x - 2)(x - 3)$
26. $x^3 - 24x^2 + 192x - 512 = (x - 8)^3$
27. $3t^3 + 4t^2 - t - 2 = 3\left(t - \frac{2}{3}\right)(t + 1)^2$

28. $2t^3 - 3t^2 - 11t + 6 = 2\left(t - \frac{1}{2}\right)(t+2)(t-3)$

29. $z^3 + 2z^2 - 3z - 6 = (z+2)(z+\sqrt{3})(z-\sqrt{3})$

30. $2z^3 - z^2 - 10z + 5 = 2\left(z - \frac{1}{2}\right)(z+\sqrt{5})(z-\sqrt{5})$

31. $4x^4 - 28x^3 + 61x^2 - 42x + 9 = 4\left(x - \frac{1}{2}\right)^2(x-3)^2$

32. $t^5 + 2t^4 - 12t^3 - 38t^2 - 37t - 12 = (t+1)^3(t+3)(t-4)$

33. $125z^5 - 275z^4 - 2265z^3 - 3213z^2 - 1728z - 324 = 125\left(z + \frac{3}{5}\right)^3(z+2)(z-6)$

34. $x^2 - 2x - 2 = (x - (1-\sqrt{3}))(x - (1+\sqrt{3}))$

35. $p(x) = 5x^2 - 6x - 4$

38.
- For $f(x) = x^3 - 2x + 1$, the coefficients $1 + (-2) + 1 = 0$ and $f(x) = (x-1)(x^2+x-1)$.
- For $f(x) = 3x^4 - x - 2$ the coefficients $3 + (-1) + (-2) = 0$ and $f(x) = (x-1)(3x^3+3x^2+3x+2)$.

39.
- $\dfrac{x-a}{x-a} = 1$
- $\dfrac{x^2-a^2}{x-a} = x + a$
- $\dfrac{x^3-a^3}{x-a} = x^2 + ax + a^2$
- $\dfrac{x^4-a^4}{x-a} = x^3 + ax^2 + a^2x + a^3$
- $\dfrac{x^5-a^5}{x-a} = x^4 + ax^3 + a^2x^2 + a^3x + a^4$

Following the pattern:

- $\dfrac{x^{10}-a^{10}}{x-a} = x^9 + ax^8 + a^2x^7 + a^3x^6 + a^4x^5 + a^5x^4 + a^6x^3 + a^7x^2 + a^8x + a^9$
- $\dfrac{x^n-a^n}{x-a} = x^{n-1} + ax^{n-2} + a^2x^{n-3} + \ldots + a^{n-2}x + a^{n-1}$

40. Put $x = 1$ and $a = r$ so that $1 + r + r^2 + \cdots + r^{n-2} + r^{n-1} = \dfrac{1-r^n}{1-r}$. Here, $r \neq 1$ as otherwise we'd be dividing by 0.

2.3 Real Zeros of Polynomials

2.4 Complex Zeros and the Fundamental Theorem of Algebra